

Transformers

Transformers are a type of neural network architecture that transforms or changes an input sequence into an output sequence. They do this by learning context and tracking relationships between sequence components. Its fundamental layer is called Attention. A typical transformer model has multiple transformer blocks stacked together. Each transformer block has two main components: a multi-head self-attention mechanism and a position-wise feed-forward neural network.

(a) Self-attention mechanism: The self-attention mechanism is the core engine of modern AI models like ChatGPT. It allows a computer to look at every word in a sentence at the same time and decide which other words are most important to its meaning. It focuses on relevant parts of the input when making predictions.

(b) Vector embedding: A vector embedding is a way to convert words, images, or ideas into a string of numbers so that a computer can understand their meaning.

(c) Feed forward Neural Network: A Feed-Forward Neural Network (FFNN) is the simplest and most foundational type of artificial neural network. In an FFNN, information moves in only one direction: forward. It starts at the input nodes, travels through hidden layers, and exits through output nodes. There are no loops or cycles in the network.